

EXPERIMENT 2

Title

Conversion of the given E-R Diagram of an Indian E-Commerce Platform into a Relational Schema and implementation using MySQL.

Aim

To convert the given E-R diagram into a relational schema and implement it in MySQL using primary keys, foreign keys, NOT NULL, UNIQUE, ON DELETE CASCADE and ON DELETE SET NULL constraints. To insert sample data and demonstrate referential integrity violations.

Theory

A relational schema represents the structure of a relational database in terms of tables, attributes and keys. Entities are converted into tables, attributes become columns, and relationships are represented using foreign keys or separate relation tables. A 1:N relationship is represented by placing the key of the one-side in the many-side. An N:M relationship requires a separate junction table. Multivalued attributes are represented using separate tables. The ISA specialization of PRODUCT is represented by ELECTRONICS, FASHION and BOOKS subtype tables.

Relational Schema

```
CUSTOMER(CustomerID, Name, Email, Phone, DOB, Street, City, State, PIN)
SELLER(SellerID, SellerName, Email, Phone)
CATEGORY(CategoryID, CategoryName, Description)
PRODUCT(ProductID, ProductName, Price, StockQty, SellerID, CategoryID)
ELECTRONICS(ProductID)
FASHION(ProductID)
BOOKS(ProductID)
PRODUCT_IMAGE(ProductID, ProductImage)
CUSTOMER_WISHLIST(CustomerID, WishlistItem)
ORDERS(OrderID, OrderDate, OrderStatus, TotalAmount, CustomerID)
ORDER_ITEM(OrderID, ProductID)
ADDRESS(AddressID, Street, City, State, PIN)
PAYMENT(PaymentID, PaymentDate, Amount, PaymentMethod, OrderID)
PAYMENT_STATUS(PaymentID, PaymentStatus)
DELIVERY(DeliveryID, DeliveryDate, DeliveryStatus, OrderID, AddressID)
```

MySQL CREATE TABLE Statements

```
CREATE TABLE CUSTOMER (
    CustomerID INT PRIMARY KEY,
    Name VARCHAR(100) NOT NULL,
    Email VARCHAR(100) NOT NULL UNIQUE,
    Phone VARCHAR(15) NOT NULL UNIQUE,
    DOB DATE, Street VARCHAR(100), City VARCHAR(50),
    State VARCHAR(50), PIN VARCHAR(10)
);

CREATE TABLE SELLER (
    SellerID INT PRIMARY KEY,
    SellerName VARCHAR(100) NOT NULL,
    Email VARCHAR(100) NOT NULL UNIQUE,
    Phone VARCHAR(15) NOT NULL UNIQUE
);
```

```

CREATE TABLE CATEGORY (
    CategoryID INT PRIMARY KEY,
    CategoryName VARCHAR(100) NOT NULL UNIQUE,
    Description VARCHAR(255)
);

CREATE TABLE PRODUCT (
    ProductID INT PRIMARY KEY,
    ProductName VARCHAR(150) NOT NULL,
    Price DECIMAL(10,2) NOT NULL,
    StockQty INT NOT NULL,
    SellerID INT, CategoryID INT,
    FOREIGN KEY (SellerID) REFERENCES SELLER(SellerID) ON DELETE SET NULL,
    FOREIGN KEY (CategoryID) REFERENCES CATEGORY(CategoryID) ON DELETE SET NULL
);

CREATE TABLE ELECTRONICS (
    ProductID INT PRIMARY KEY,
    FOREIGN KEY (ProductID) REFERENCES PRODUCT(ProductID) ON DELETE CASCADE
);

CREATE TABLE FASHION (
    ProductID INT PRIMARY KEY,
    FOREIGN KEY (ProductID) REFERENCES PRODUCT(ProductID) ON DELETE CASCADE
);

CREATE TABLE BOOKS (
    ProductID INT PRIMARY KEY,
    FOREIGN KEY (ProductID) REFERENCES PRODUCT(ProductID) ON DELETE CASCADE
);

CREATE TABLE PRODUCT_IMAGE (
    ProductID INT NOT NULL,
    ProductImage VARCHAR(255) NOT NULL,
    PRIMARY KEY (ProductID, ProductImage),
    FOREIGN KEY (ProductID) REFERENCES PRODUCT(ProductID) ON DELETE CASCADE
);

CREATE TABLE CUSTOMER_WISHLIST (
    CustomerID INT NOT NULL,
    WishlistItem VARCHAR(255) NOT NULL,
    PRIMARY KEY (CustomerID, WishlistItem),
    FOREIGN KEY (CustomerID) REFERENCES CUSTOMER(CustomerID) ON DELETE CASCADE
);

CREATE TABLE ORDERS (
    OrderID INT PRIMARY KEY,
    OrderDate DATE NOT NULL,
    OrderStatus VARCHAR(50) NOT NULL,
    TotalAmount DECIMAL(10,2) NOT NULL,
    CustomerID INT NOT NULL,
    FOREIGN KEY (CustomerID) REFERENCES CUSTOMER(CustomerID) ON DELETE CASCADE
);

CREATE TABLE ORDER_ITEM (
    OrderID INT NOT NULL,
    ProductID INT NOT NULL,
    PRIMARY KEY (OrderID, ProductID),
    FOREIGN KEY (OrderID) REFERENCES ORDERS(OrderID) ON DELETE CASCADE,
    FOREIGN KEY (ProductID) REFERENCES PRODUCT(ProductID) ON DELETE CASCADE
);

CREATE TABLE ADDRESS (
    AddressID INT PRIMARY KEY,

```

```

        Street VARCHAR(100) NOT NULL,
        City VARCHAR(50) NOT NULL,
        State VARCHAR(50) NOT NULL,
        PIN VARCHAR(10) NOT NULL
    );

CREATE TABLE PAYMENT (
    PaymentID INT PRIMARY KEY,
    PaymentDate DATE NOT NULL,
    Amount DECIMAL(10,2) NOT NULL,
    PaymentMethod VARCHAR(50) NOT NULL,
    OrderID INT NOT NULL,
    FOREIGN KEY (OrderID) REFERENCES ORDERS(OrderID) ON DELETE CASCADE
);

CREATE TABLE PAYMENT_STATUS (
    PaymentID INT NOT NULL,
    PaymentStatus VARCHAR(50) NOT NULL,
    PRIMARY KEY (PaymentID, PaymentStatus),
    FOREIGN KEY (PaymentID) REFERENCES PAYMENT(PaymentID) ON DELETE CASCADE
);

CREATE TABLE DELIVERY (
    DeliveryID INT PRIMARY KEY,
    DeliveryDate DATE,
    DeliveryStatus VARCHAR(50) NOT NULL,
    OrderID INT NOT NULL,
    AddressID INT,
    FOREIGN KEY (OrderID) REFERENCES ORDERS(OrderID) ON DELETE CASCADE,
    FOREIGN KEY (AddressID) REFERENCES ADDRESS(AddressID) ON DELETE SET NULL
);

```

Sample Data

```

INSERT INTO CUSTOMER VALUES
(1, 'Rahul Sharma', 'rahul@gmail.com', '9876543210', '2003-05-15', 'MG Road', 'Delhi', 'Delhi', '110001'),
(2, 'Priya Singh', 'priya@gmail.com', '9876543211', '2002-08-20', 'Park Street', 'Kolkata', 'West Bengal', '700016');

INSERT INTO SELLER VALUES
(101, 'Tech World', 'tech@gmail.com', '9000000001'),
(102, 'Fashion Hub', 'fashion@gmail.com', '9000000002');

INSERT INTO CATEGORY VALUES
(201, 'Electronics', 'Electronic products'),
(202, 'Fashion', 'Clothing and accessories'),
(203, 'Books', 'Books and study material');

INSERT INTO PRODUCT VALUES
(301, 'Laptop', 65000.00, 10, 101, 201),
(302, 'T-Shirt', 999.00, 50, 102, 202),
(303, 'DBMS Book', 599.00, 20, 101, 203);

INSERT INTO ELECTRONICS VALUES (301);
INSERT INTO FASHION VALUES (302);
INSERT INTO BOOKS VALUES (303);

INSERT INTO PRODUCT_IMAGE VALUES
(301, 'laptop1.jpg'), (301, 'laptop2.jpg'), (302, 'tshirt.jpg');

INSERT INTO CUSTOMER_WISHLIST VALUES
(1, 'Gaming Laptop'), (1, 'Smartphone'), (2, 'T-Shirt');

INSERT INTO ORDERS VALUES
(1001, '2026-08-20', 'Confirmed', 65999.00, 1),

```

```

(1002,'2026-08-21','Pending',999.00,2);

INSERT INTO ORDER_ITEM VALUES
(1001,301),(1001,303),(1002,302);

INSERT INTO ADDRESS VALUES
(501,'MG Road','Delhi','Delhi','110001'),
(502,'Park Street','Kolkata','West Bengal','700016');

INSERT INTO PAYMENT VALUES
(601,'2026-08-20',50000.00,'UPI',1001),
(602,'2026-08-20',15999.00,'Credit Card',1001),
(603,'2026-08-21',999.00,'UPI',1002);

INSERT INTO PAYMENT_STATUS VALUES
(601,'Successful'),(602,'Successful'),(603,'Pending');

INSERT INTO DELIVERY VALUES
(701,'2026-08-22','Delivered',1001,501),
(702,'2026-08-23','Processing',1001,502),
(703,NULL,'Processing',1002,502);

```

Referential Integrity Violations

```

-- Invalid CustomerID
INSERT INTO ORDERS VALUES (1003,'2026-08-25','Pending',5000.00,999);

-- Invalid ProductID
INSERT INTO ORDER_ITEM VALUES (1003,999);

-- Invalid SellerID
INSERT INTO PRODUCT VALUES (304,'Mobile Phone',25000.00,15,999,201);

-- Invalid CategoryID
INSERT INTO PRODUCT VALUES (305,'Headphones',2000.00,25,101,999);

-- Invalid OrderID
INSERT INTO PAYMENT VALUES (604,'2026-08-25',5000.00,'UPI',9999);

-- Invalid AddressID
INSERT INTO DELIVERY VALUES (704,NULL,'Processing',1002,9999);

```

ON DELETE CASCADE Demonstration

```

DELETE FROM ORDERS WHERE OrderID = 1001;

SELECT * FROM ORDER_ITEM WHERE OrderID = 1001;
SELECT * FROM PAYMENT WHERE OrderID = 1001;
SELECT * FROM DELIVERY WHERE OrderID = 1001;

```

The related ORDER_ITEM, PAYMENT and DELIVERY records for OrderID 1001 are automatically deleted because their foreign keys use ON DELETE CASCADE.

ON DELETE SET NULL Demonstration

```

DELETE FROM SELLER WHERE SellerID = 101;
SELECT ProductID, ProductName, SellerID FROM PRODUCT;

DELETE FROM ADDRESS WHERE AddressID = 501;
SELECT * FROM DELIVERY;

```

When SellerID 101 is deleted, the products remain and SellerID becomes NULL. When AddressID 501 is deleted, the delivery record remains and AddressID becomes NULL.

Result

The given E-R diagram was successfully converted into a relational schema and implemented in MySQL. Primary keys, foreign keys, NOT NULL, UNIQUE, ON DELETE CASCADE and ON DELETE SET NULL constraints were applied. Sample data was inserted successfully, and referential integrity violations were demonstrated.

Conclusion

Thus, the Indian E-Commerce Platform database was successfully designed and implemented in MySQL while maintaining referential integrity and enforcing the relationships represented in the E-R diagram.